

Phonemic Analysis

Dr. Josef Fruehwald

```
source(here::here("_defaults.R"))
```

```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ lubridate 1.9.5     ✓ tibble    3.3.1
✓ purrr     1.2.2     ✓ tidyr     1.3.2
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tinytable)
library(rlang)
```

```

Attaching package: 'rlang'

The following objects are masked from 'package:purrr':

    flatten, flatten_chr, flatten_dbl, flatten_int, flatten_lgl,
    flatten_raw, invoke, splice
```



```
library(patchwork)
```

Recap: Phonetics vs Phonology

Phonetics

“pin”	[pʰɪn]
“spin”	[spɪn]
“bin”	[pɪn]

One has a long-lag VOT and the other two have short-lag VOT

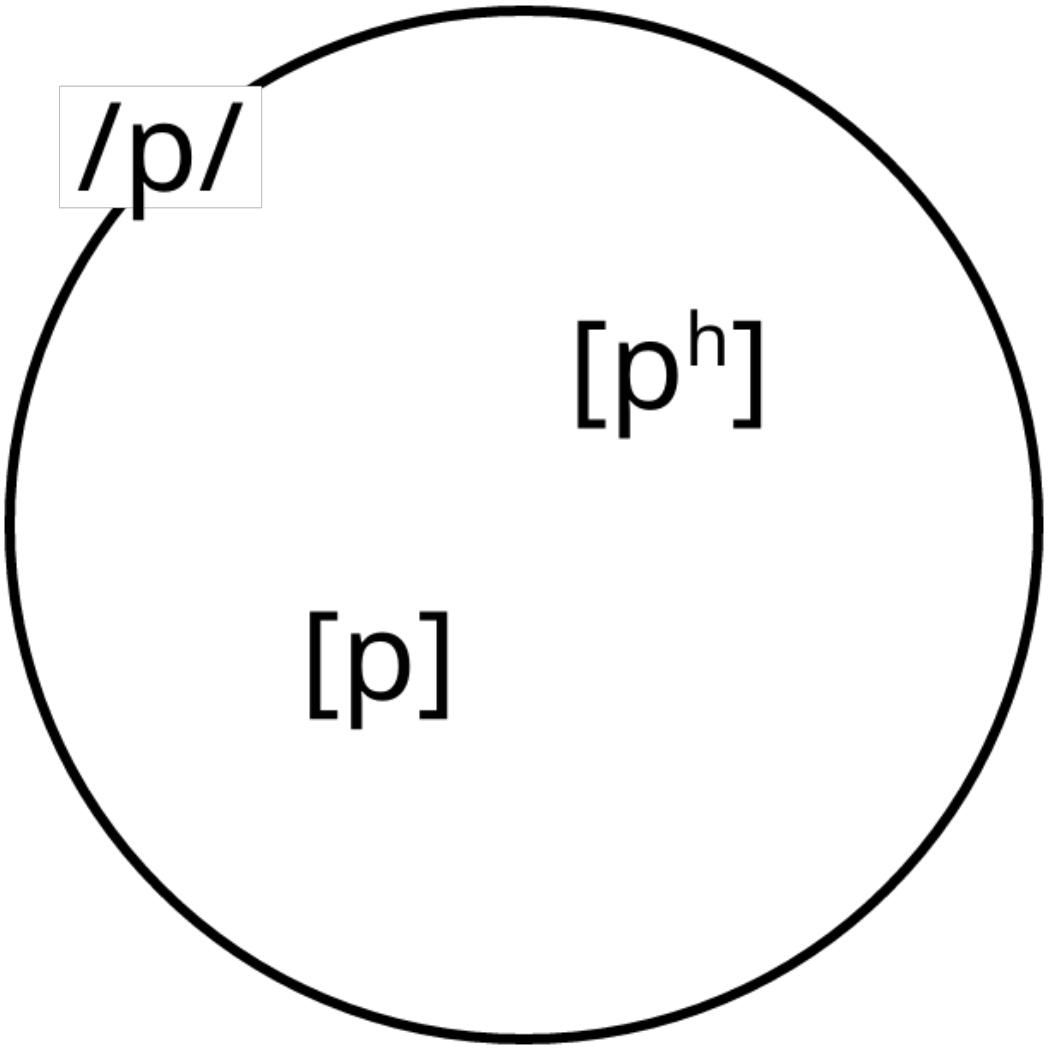
Phonology

“pin”	/pɪn/
“spin”	/spɪn/
“bin”	/bɪn/

The first two are allophones of /p/, and the third is an allophone of /b/

Recap: Phoneme

> Phoneme: A label for a set of sounds that all count as basically the same.



The phoneme /p/ has at least two *allophones*, [p] and [pʰ]

Recap: Diagnostics

Minimal Pairs

If there are at least two words that are differentiated by just these two sounds, they are *contrastive*.

English

[d]

“dough”	doʊ
“den”	dɛn

[ð]

“though”	ðoʊ
“then”	ðɛn

Recap: Diagnostics

Complementary Distribution

If two sounds never appear in the same position, they are probably *allophonic*

Spanish

Word initial

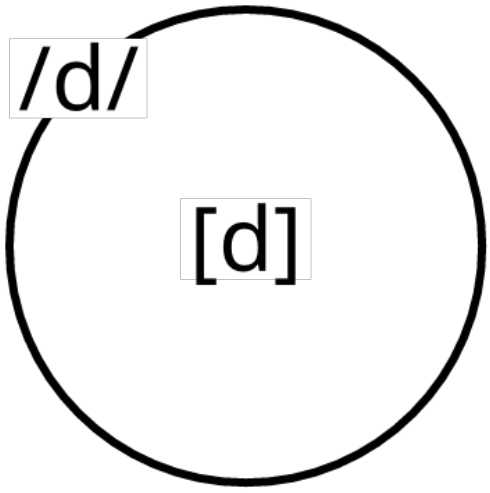
lady	“dama”	dɑmɑ

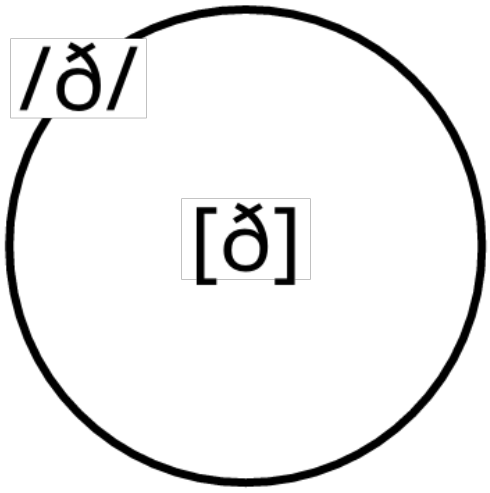
Elsewhere

side	“lado”	lɑðo

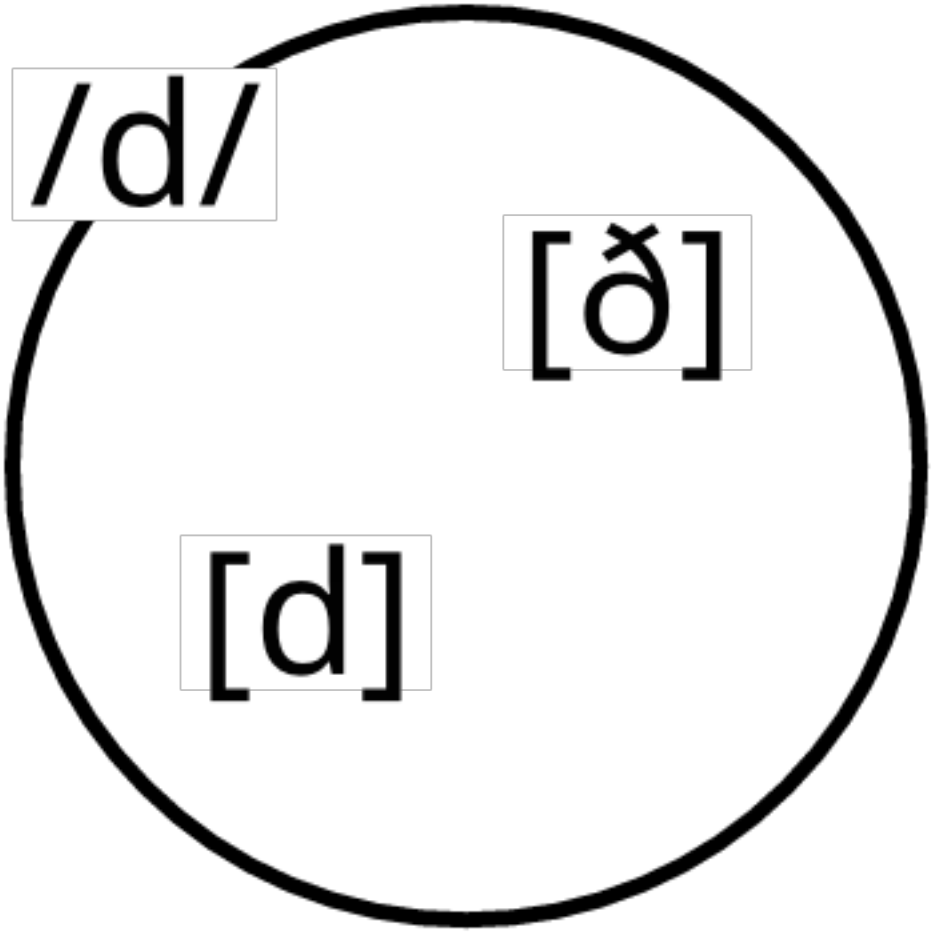
Recap: Diagnostics

English:





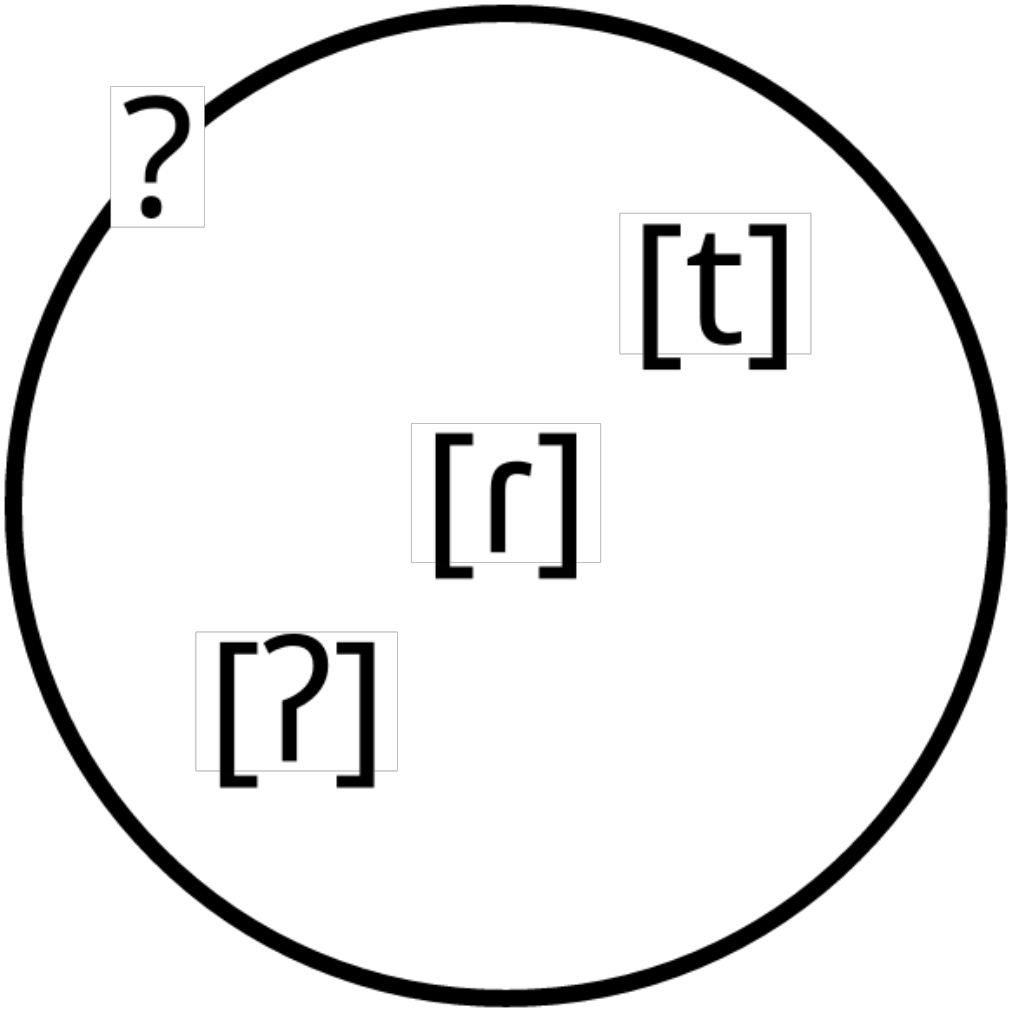
Spanish:



Phonemic Analysis

What is the underlying form?

sit	sɪt
sitting	sɪɾɪŋ
sittin'	sɪʔn̩

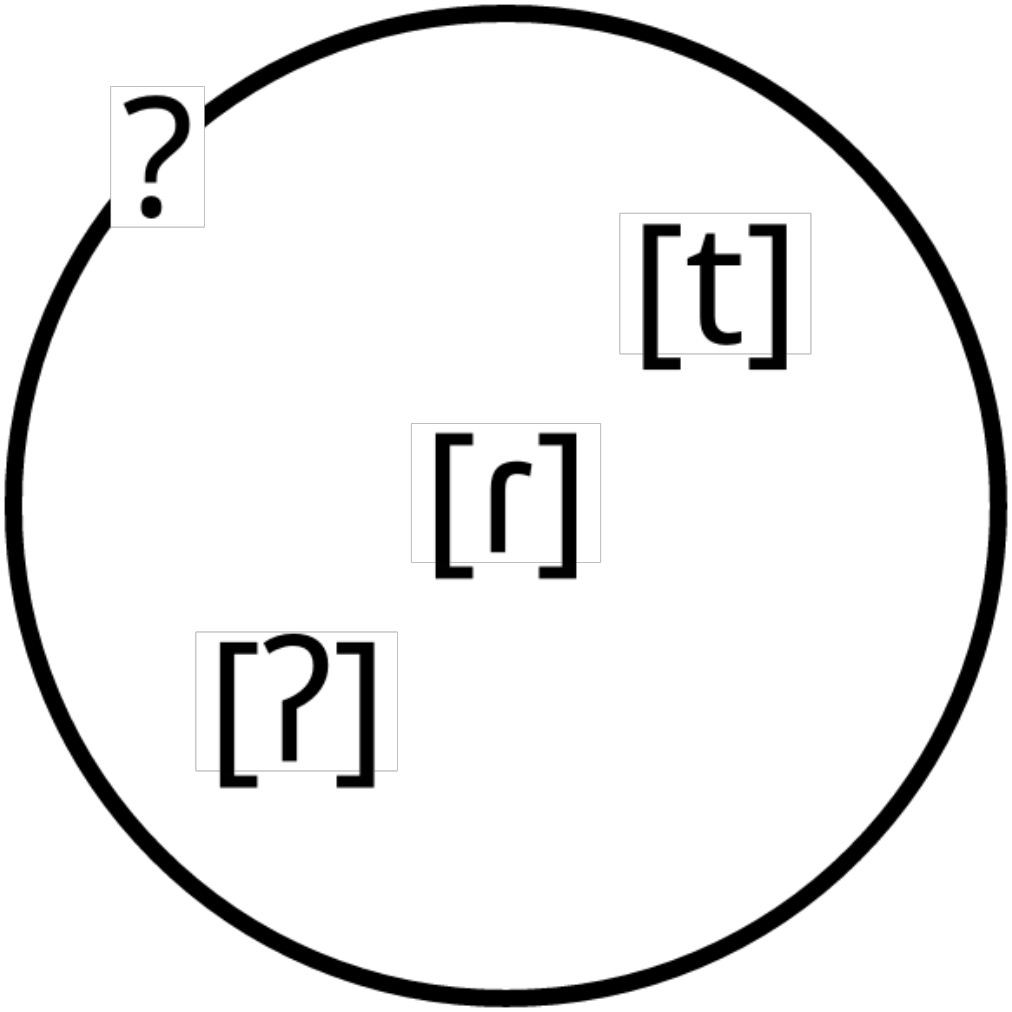


Phonemic Analysis

What is the underlying form?

Choice of the underlying form here will indicate our thought about whether the phoneme is

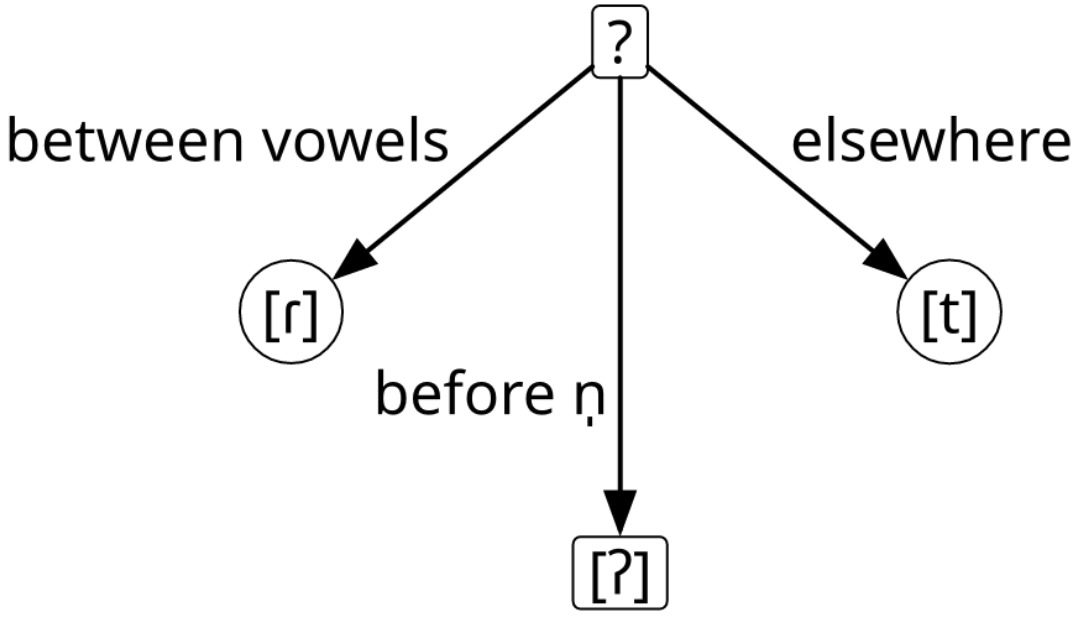
- Voiced or Voiceless
- A Stop or a Flap
- Coronal or Non-Coronal



Phonemic Analysis

What is the underlying form?

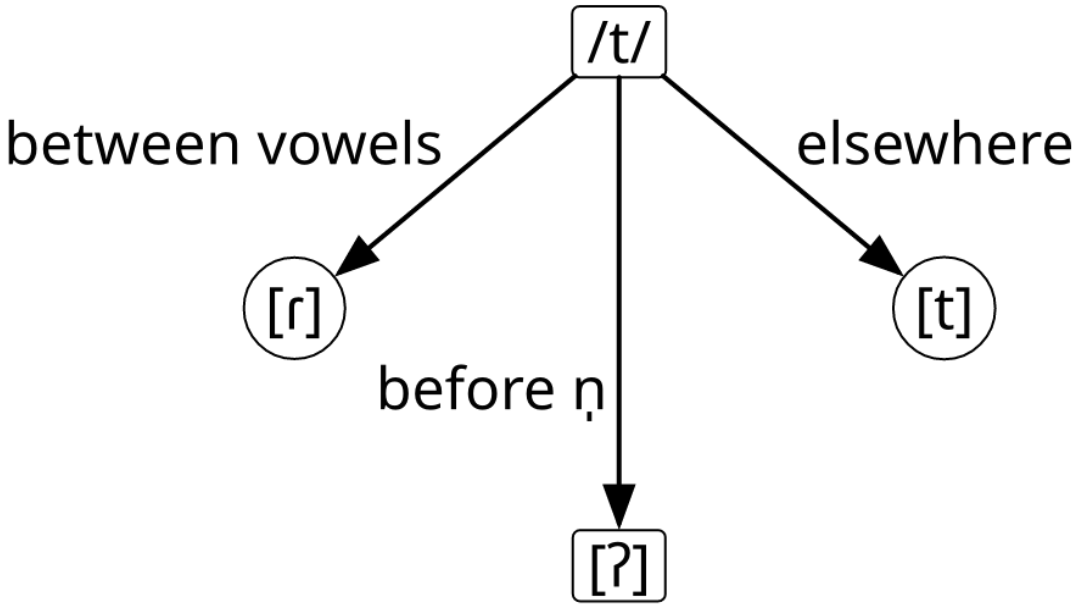
Which form shows up as the default?



Phonemic Analysis

What is the underlying form?

Which form shows up as the default?





Phonemic Analysis

What is the underlying form?

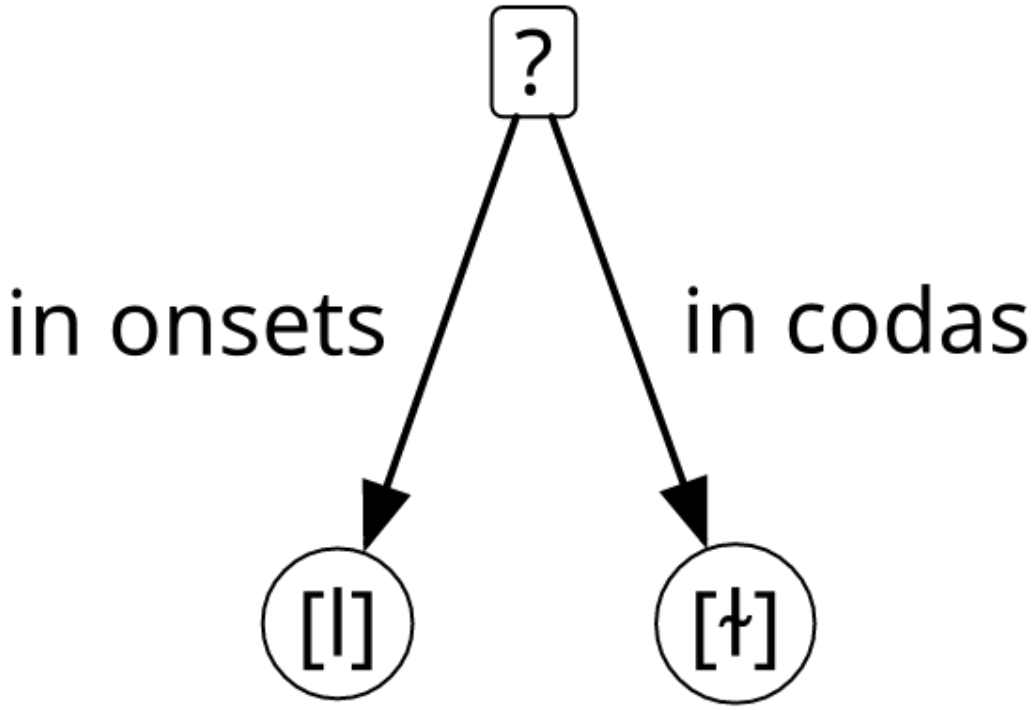
```
tribble(
  ~transcription, ~word,
  !!!str_split("lif leaf
   lup loop
   lɛd lead
   leɪm lame
   bleɪm blame",
   "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
tribble(
  ~transcription, ~word,
  !!!str_split(
    "fiɬ feel
     puɬ pool
     dɛɬ dell
     meɪɬ mail
     hɛɬp help",
   "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

lif	*leaf*
lup	*loop*
lɛd	*lead*
leɪm	*lame*
bleɪm	*blame*

fiɬ	*feel*
puɬ	*pool*
dɛɬ	*dell*
meɪɬ	*mail*
hɛɬp	*help*

Phonemic Analysis

Neither onsets nor codas are more “basic”



Neutralization

```
tribble(
  ~transcription, ~word,
  !!!str_split("bɛt bet
pæt pat
ɹəsaɪt recite
bɛd bed
pæd pad
```

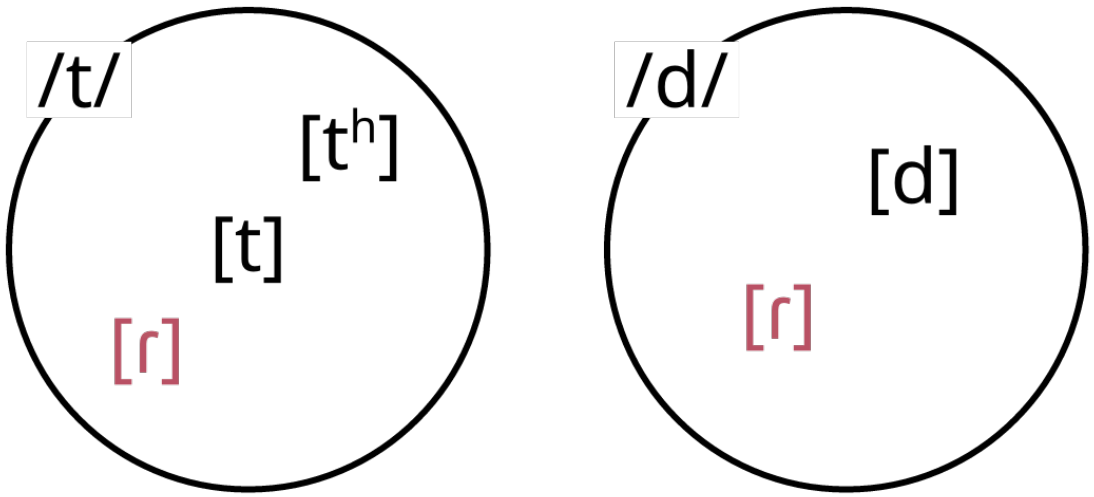


```
bɹaɪd bride",
    "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
tribble(
  ~transcription, ~word,
  !!!str_split(
  "bɛɾɪŋ betting
pæɾɪŋ patting
ɹəsaɪɾ̩ recital
bɛɾɪŋ bedding
pæɾɪŋ padding
bɹaɪɾ̩ bridal",
    "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

bɛt	*bet*
pæt	*pat*
ɹəsaɪt	*recite*
bɛd	*bed*
pæd	*pad*
bɹaɪd	*bride*

bɛɾɪŋ	*betting*
pæɾɪŋ	*patting*
ɹəsaɪɾ̩	*recital*
bɛɾɪŋ	*bedding*
pæɾɪŋ	*padding*
bɹaɪɾ̩	*bridal*

Neutralization



Neutralization - Consequences

What evidence could you come up with for the underlying forms of these words?

```
tribble(
  ~transcription, ~word,
  !!!str_split(
  "taɹ tidal
vaɹ vital
taɹ title
spaɹ spider",
   "\\s+"
  )[[1]]
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

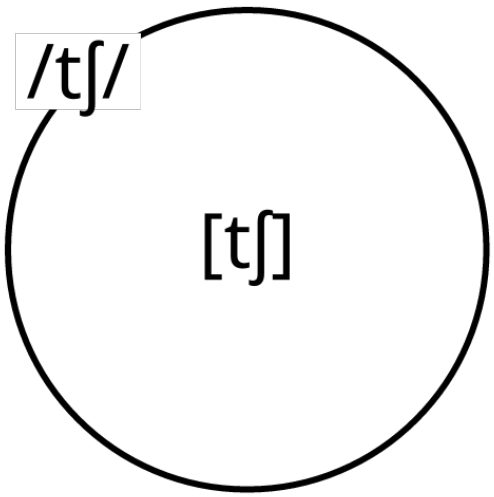
taɪɾl̩	*tidal*
vaɪɾl̩	*vital*
taɪɾl̩	*title*
spaɪɾɹ̩	*spider*

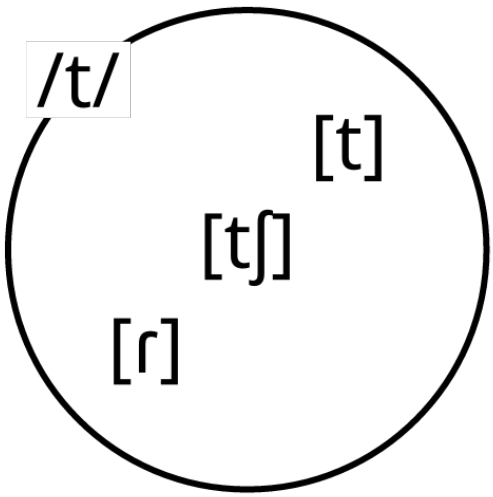
Neutralization

```
tribble(
  ~transcription, ~word,
  !!!str_split(
  "bɛt bet
bɛtʃju bet-you
bætʃ batch
lɛd lead
lɛdʒju lead-you
lɛdʒ ledge",
   "\\s+"
  )[[1]]
) |>
  mutate(
    word = str_replace(word, pattern = "-", replacement = " ")
  ) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 2,
    italic = T
  )
```

bɛt	*bet*
bɛtʃju	*bet you*
bætʃ	*batch*
lɛd	*lead*
lɛdʒju	*lead you*
lɛdʒ	*ledge*

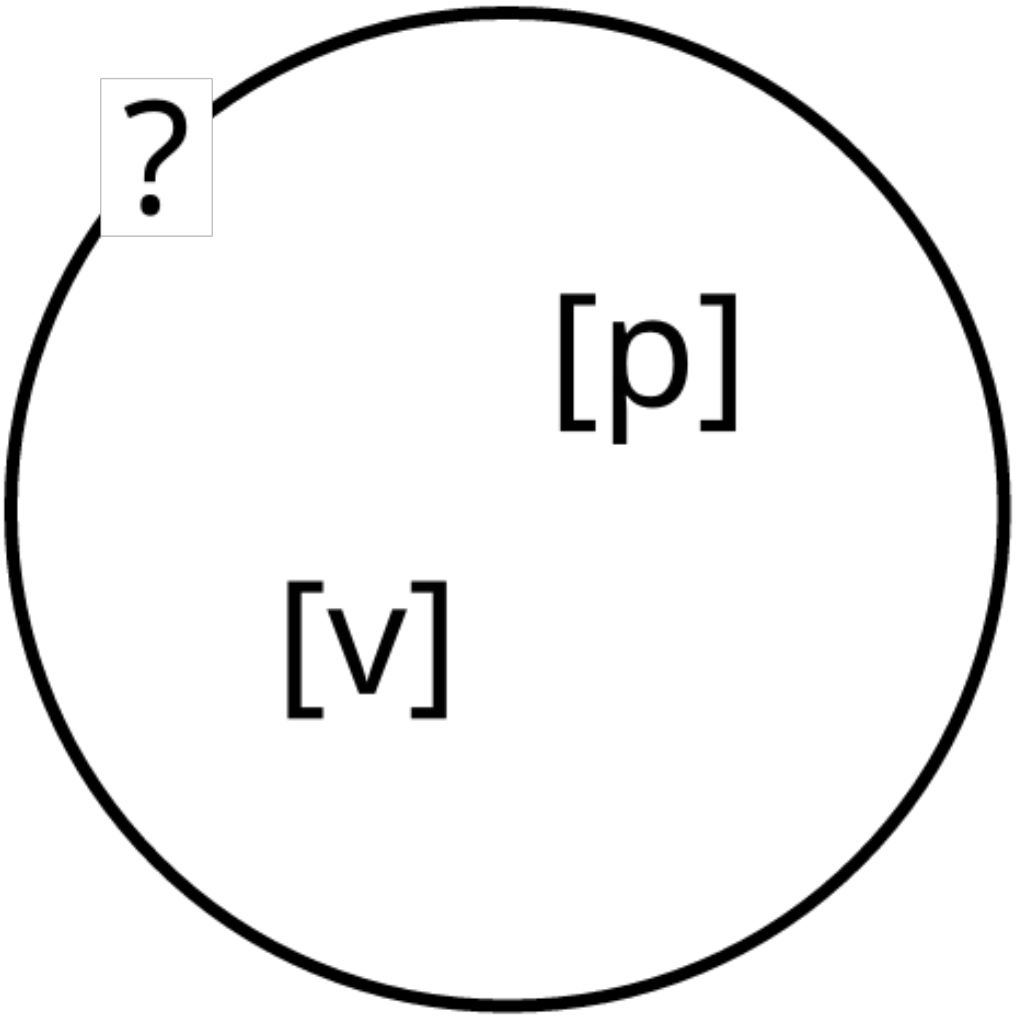
Neutralization





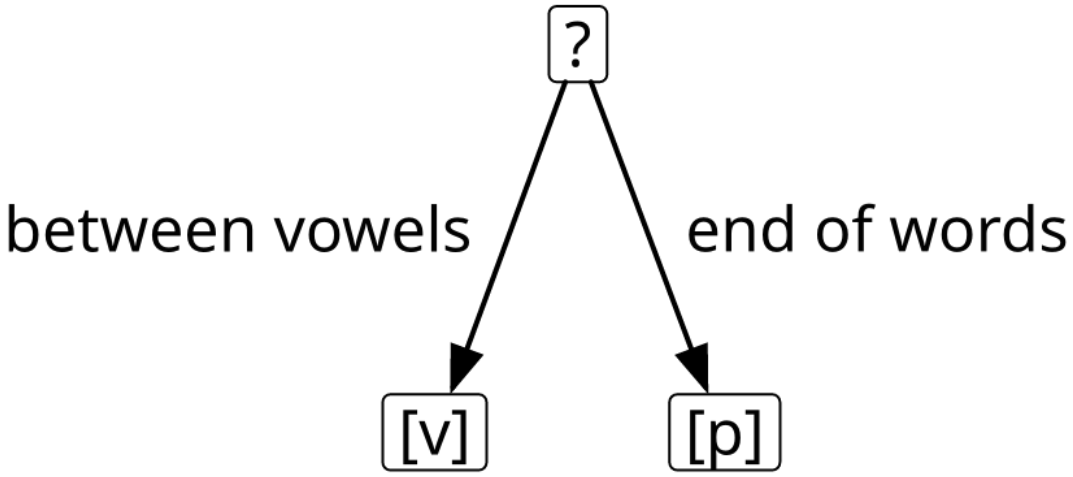
Abstractness

ʃovɪç	*Schwowisch (a German dialect)*
ʃvop	*a speaker of Schwowisch*



Abstractness

ʃvovɪç	*Schwowisch (a German dialect)*
ʃvop	*a speaker of Schwowisch*



Abstractness

ʃvovɪç *Schwowisch (a German dialect)*

ʃvop *a speaker of Schwowisch*

